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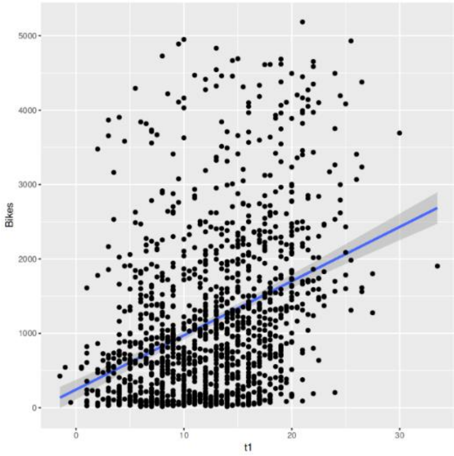
Applied Data Analysis Regression Project:

Part A) Regression modelling

1. Visualising the relationships between the dependent variable Bikes (No. bikes hired) and the variables; Season, Hour, Rain and t1.

For the continuous variable t1 (Temperatures in Degrees Celsius), a scatterplot with t1 on the x axis and Bikes on the y-axis should be sufficient to ascertain whether there is a relationship.

There doesn't seem to be much of a correlation between bikes hired and temperature, the regression line does not look to be good fit.

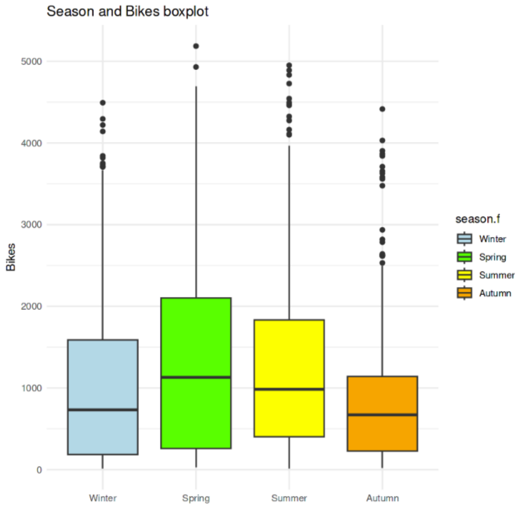


Moving on, for the categorical variables boxplots should be sufficient to visualise a relationship between the variables.

Seasons and Bikes Hired:

Here we can see there is some correlation between the seasons and bikes hired. The median bikes hired are highest in the Spring, next highest in the Summer, and are about even for Autumn and the Winter, the two with the lowest medians.

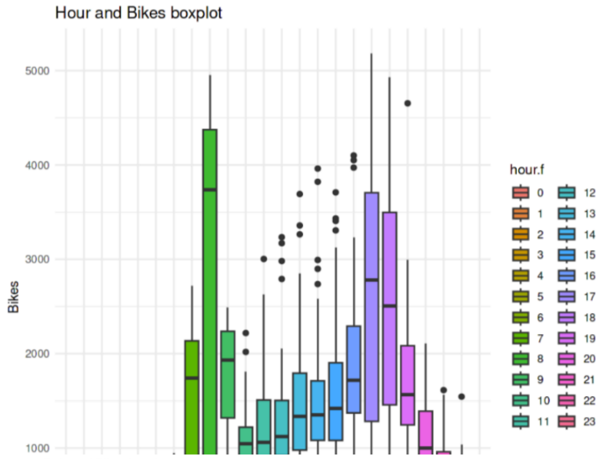
However, there are certainly differences in the distributions, both the winter and summer have skewed distributions, with the most variability in the Spring, an approximately equal amount in the Summer and Winter and the least variability in the Autumn.



Hours/Time and Bikes Hired:

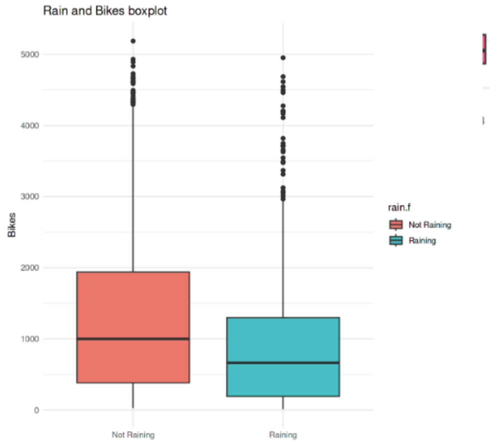
The boxplot for hours (time) and bikes hired indicates a relationship between the time of day and the amount of bikes hired. In the late night the medians are low, approximately 0 for some hours, then in the morning the median bikes hired increases and then decreases again in midday. There is then a slow growth and increase around 17,18, with a sharp decrease from 19 to 23.

There appears to be a causal relationship between the time of day and the number of bikes hired.



Rain and Bikes Hired:

There appears to be some relationship between whether it's raining or not and bikes hired, there is greater variance in bikes hired when it's not raining compared to when it is raining. As well as that the median bikes hired is higher when it is not raining compared to when it is raining.



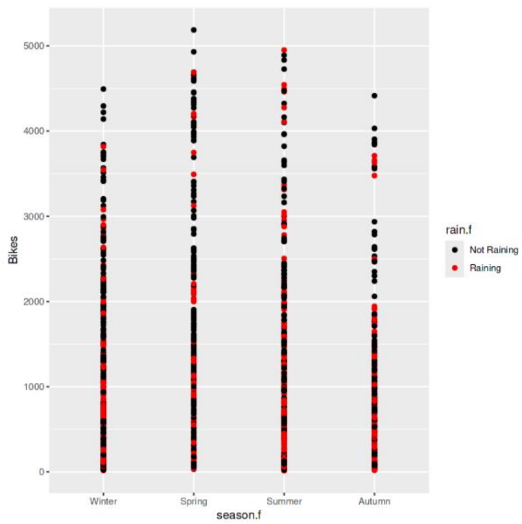
Interaction Terms:

We can hypothesize that in the warmer seasons rain may be less of a deterrence to riding a bike, so there may be an interaction between the two. Similarly as the sun rises earlier people may be more likely to rent a bike in the early morning in the summer and later at night, however in the winter the opposite may be true.

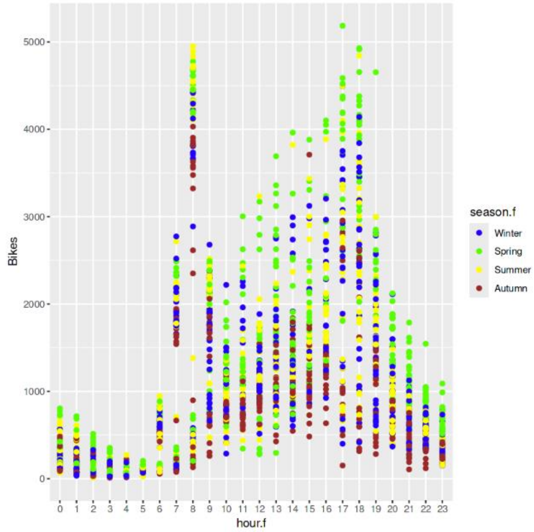
Season and Rain:

Intuitively, in the summer rain may not deter me from renting a bike, whereas in the winter it will. This sparks the possibility of an interaction between the two.

However, plotting the bikes hired in each season, highlighting the times it was raining and the times it was not shows that there is not much of an interaction. The Not Raining points are clustered evenly alongside the Raining points.



Hour and Season:



Here I can hypothesise an interaction quite easily, people may be more likely to rent a bike in the early morning and late evening in the summer/spring when it's still light out, whereas they could be less likely in the Winter months.

However from the plot we can just see that generally more bikes are hired in the Spring/Summer months and less bikes hired in the Autumn/Winter months. But, it's a general rule not specific to the time of day. Therefore I do not think there is an interaction between the two.

Conclusion:

Of the four variables, our plots indicate that hour is the most predictive, with season and rain as close seconds. The temperature variable is not predictive.

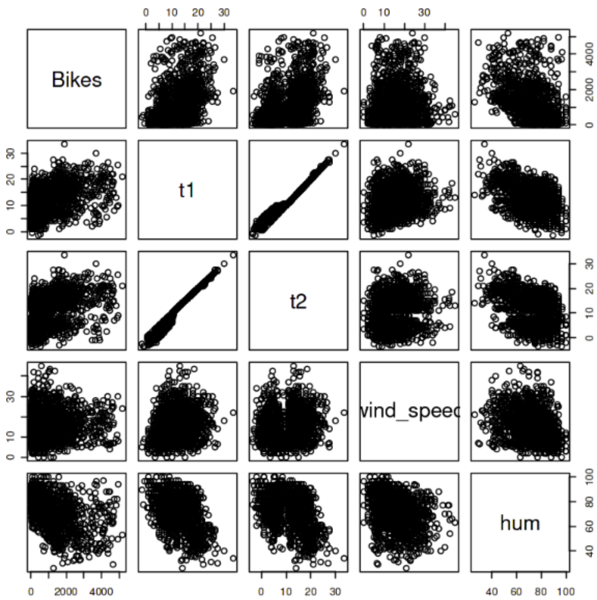
2. Finding a suitable regression model for the relationship between bikes hired and the other variables in the data set as potential predictors.

Checking the relationships of the other variables:

Clearly hour is the most predictive and it is categorical, to avoid additional complexity and difficulty interpreting the final model I opted not to include the months variable in my analysis. I believe the season variable provides sufficient information and simplifies the interpretation.

Using the pairs() function I can plot a scatterplot of all the continuous variables related to the bikes hired variable.

We can see here that t1 and t2 are collinear so only one can be included in the model. We can also see a relationship between humidity and bikes, there is a negative linear relationship, the more humidity the less bikes hired and vice versa.



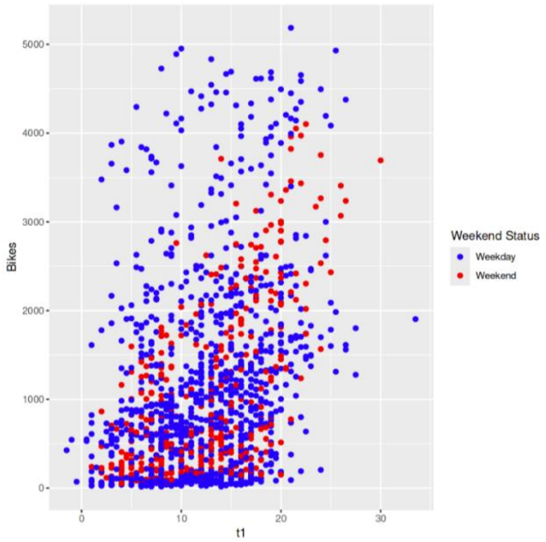
Boxplots for the holiday and weekend variables don't indicate that there is much of a relationship. To avoid overfitting I'll leave them out as well. They can be found in Appendix A.

Interaction terms: We're still lacking a lot of predictors, currently we have only humidity and hour. Using the additional variables we'll visually check for the following interactions,

- **is_weekend:t1**, during the weekday, people may always rent a bike out of necessity, however, on the weekend they may only do it if the temperature is nice.
- **is_holiday:t1**, similarly if it's a warm holiday one might rent a bike, but if it's cold they may prefer to stay in and watch Netflix.
- **is_weekend:hum**, this is a similar reasoning to is_weekend:t1, during the weekday, people may rent out of necessity, however if the weather is suitable to exercising (low humidity), they may go for a bike ride on the weekend.
- **is_weekend:wind_speed**, similar reasoning to the above.

is_weekend:t1

Here we can see that there does seem to be an interaction between temperature and the weekend term, during the weekday the number of bikes hired is fairly random and independent of temperature. However, on the weekend there is a linear relationship between temperature and bikes hired, the higher the temperature the more bikes hired.

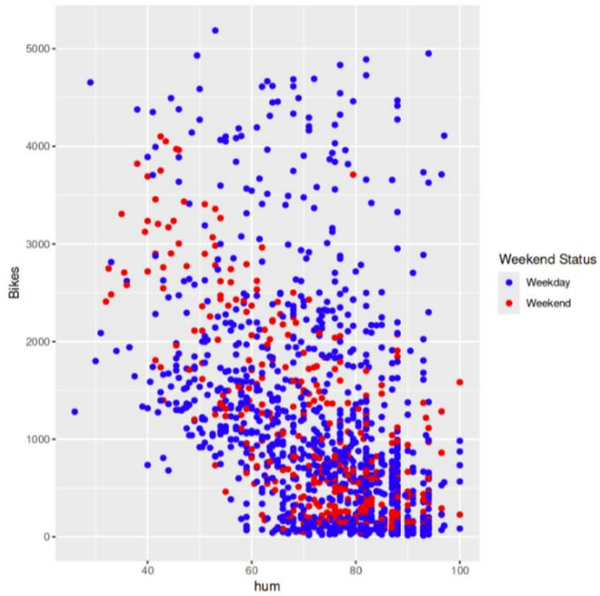


is_holiday:t1

In my dataset there were so few datapoints for is_holiday = Yes, so I could not include it in my model. The plot is found in Appendix A.

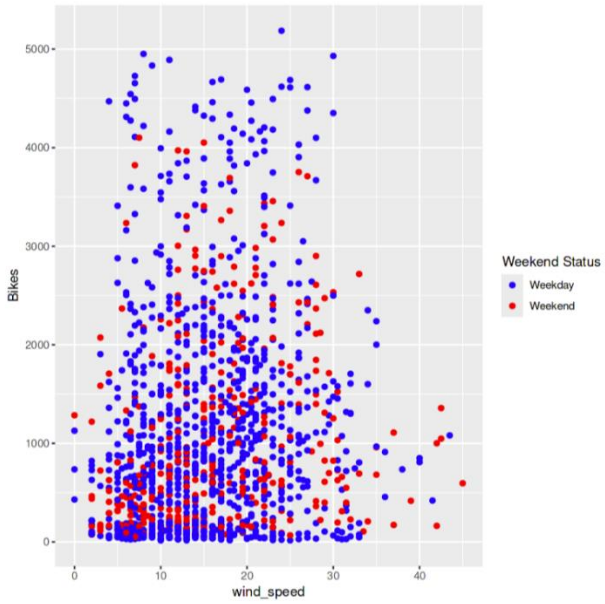
is_weekend:hum

In this case there doesn't seem to be much of a relationship on the weekday, however, on the weekend there is a negative linear relationship between humidity and bikes hired. On the weekend, the higher the humidity, the less bikes hired. On the weekdays, the humidity level doesn't seem to have a strong effect on the number of bikes hired. On the weekend however, lower humidity leads to more bikes hired and higher humidity to less bikes hired.



is_weekend:wind_speed

In contrast, there does not seem to be a relationship between the wind speed and weekend status. The data points are fairly evenly distributed regardless of weekend status.



Initial model, and initial conclusions:

Which variables are we including?

From our visual analysis we can conclude that the initial model should be made up of the variables hour.f, season.f, rain.f, and the interaction terms is_weekend:hum and is_weekend:t1.

Fitting the model as,

```
lm_v1 <- lm(Bikes ~ hum + season.f + rain.f + hour.f + is_weekend.f:hum +
is_weekend.f:t1, data = train)
```

The Anova() function returns,

A anova: 7 × 5

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
hum	1	367035478.68	367035478.68	1007.1143176	4.308951e-166
season.f	3	36486229.73	12162076.58	33.3717097	8.134523e-21
rain.f	1	39996.41	39996.41	0.1097468	7.404834e-01
hour.f	23	714482793.51	31064469.28	85.2382770	1.542441e-244
hum:is_weekend.f	1	21137808.35	21137808.35	58.0003587	4.864550e-14
is_weekend.f:t1	2	37805327.01	18902663.50	51.8673103	1.938690e-22
Residuals	1368	498557637.47	364442.72	NA	NA

This indicates that all of the variables are statistically significant, and using the summ() function for the model reveals that the VIF value for the interaction term is_weeked.f:t1 is approximately 10. This output can be found in Appendix A. For this reason, the variable was removed and the model retrained.

This is the output for an ANOVA of the retrained model with type III sums of squares.

	Sum Sq	Df	F value	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	39949375	1	102.04031	3.447589e-23
hum	31480627	1	80.40909	9.748656e-19
season.f	36375335	3	30.97045	2.268868e-19
rain.f	6842388	1	17.47710	3.092284e-05
hour.f	726141780	23	80.64090	3.586401e-235
hum:is_weekend.f	21137808	1	53.99105	3.447613e-13
Residuals	536362964	1370	NA	NA

This indicates that all of the variables are statistically significant and as the VIF values are all less than 2.5 no further retraining of the model is required.

Partial F-Test:

To assess whether the interaction between humidity and the weekend is statistically significant I will perform a partial F-test. The model without an interaction includes the variables hum, season.f, rain.f, and hour.f.

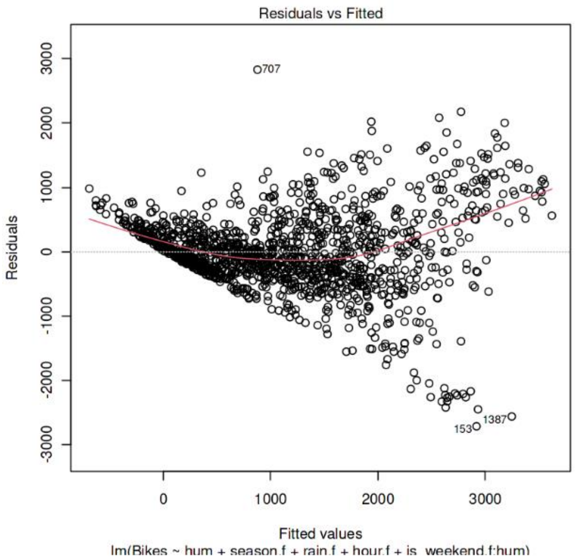
The partial F-Test gives the following output:

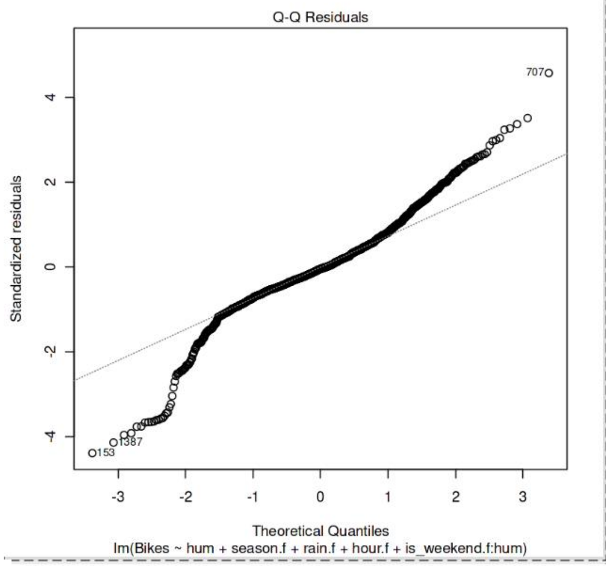
```
Analysis of Variance Table

Model 1: Bikes ~ hum + season.f + rain.f + hour.f
Model 2: Bikes ~ hum + season.f + rain.f + hour.f + hum:is_weekend.f
  Res.Df       RSS Df Sum of Sq      F    Pr(>F)
1   1371 557500773
2   1370 536362964  1  21137808 53.991 3.448e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

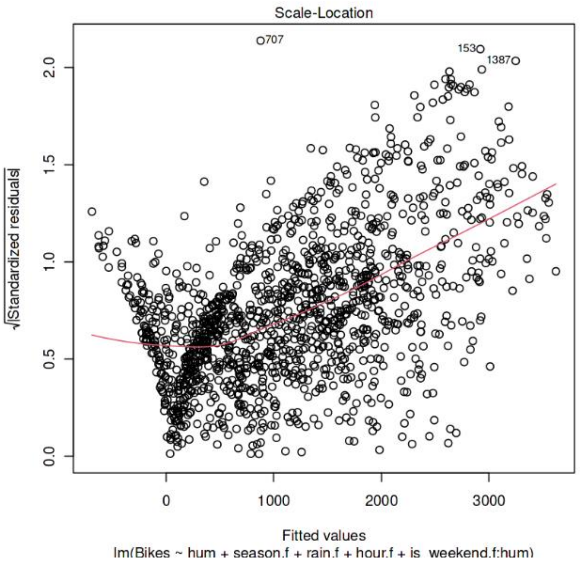
The interaction term is statistically significant and contributes 22, 137, 808 to the model’s total sum of squares, indicating a meaningful contribution to the variability in the number of bikes hired.

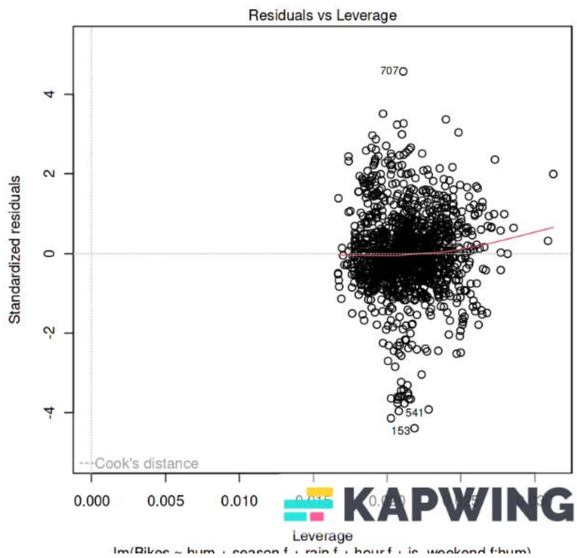
Residual Diagnostics:

Now that we have our model we will assess the residuals to establish whether it is a good model. Plotting the model gives the following output.

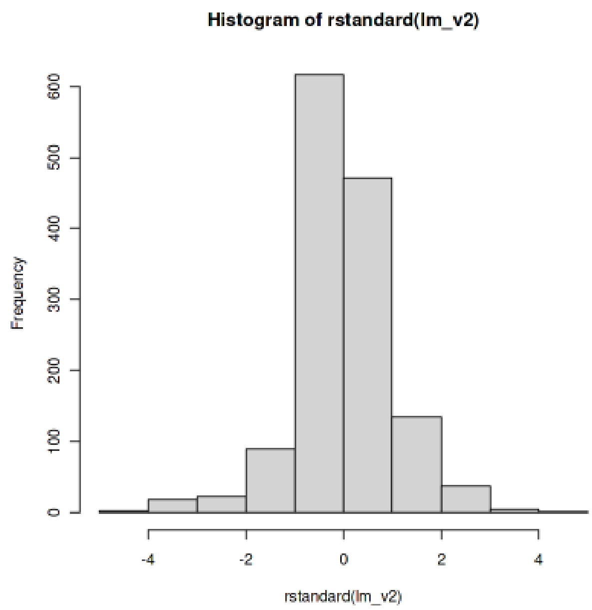






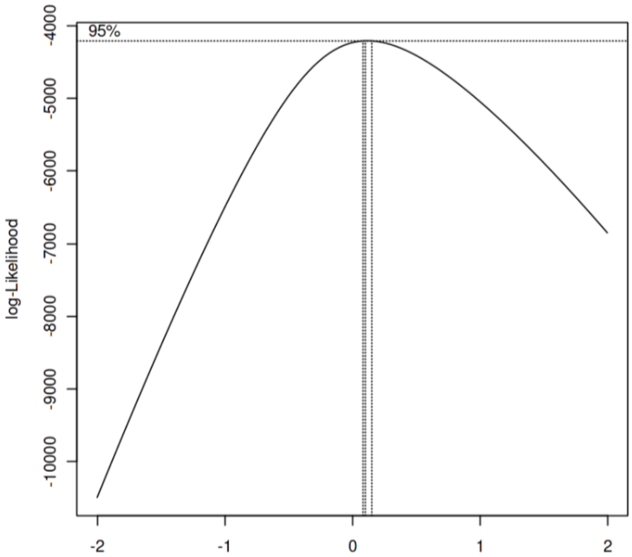


The four plots all indicate heteroscedasticity, however, the Residuals vs Leverage plot also shows that there are no influential outliers as none of the observations lie outside of Cook's dashed lines



A histogram of the residuals indicates that there is some right skewedness.

To address this concern, I applied the Box-Cox transformation using the maximum likelihood estimation to determine the optimal lambda value, which was calculated to be 0.1010. That plot can be seen on the right.

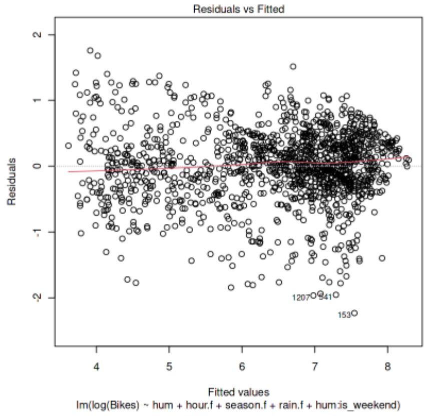


However, the difference in log-likelihood between the lambda value of 0.1010 and 0 (corresponding to a log transform) is minimal. Due to this, and the difficulty of interpreting a model with a Box-Cox transformation of 0.1010, I opted to use the log transform instead.

The transformed model was fitted with independent variable log(bikes) and dependent variables humidity (hum), season, hour, and the interaction between weekend and humidity. Now we reassess the residuals .

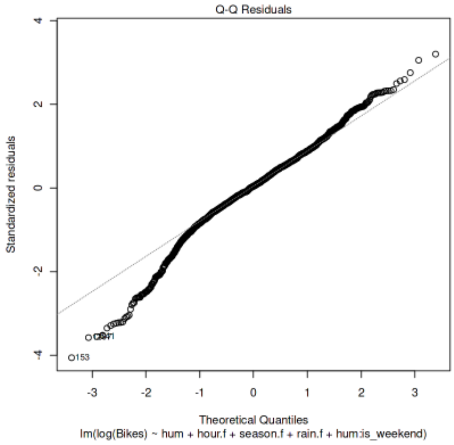
Residuals vs Fitted Plot:

The red line across the centre of the plot is roughly horizontal, indicating that the residuals are normally distributed with constant variance. This provides evidence that the residuals are normally distributed.



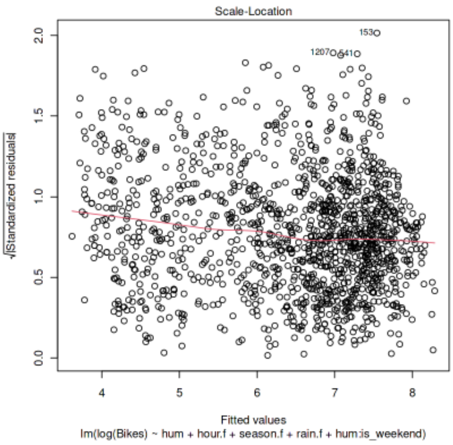
Standardized QQ-Plot:

Similar to the original model prior to transform, the QQ-plot provides the strongest evidence against the normality of the residuals assumption. The points roughly follow the line however there is some deviation at the right and left endpoints.



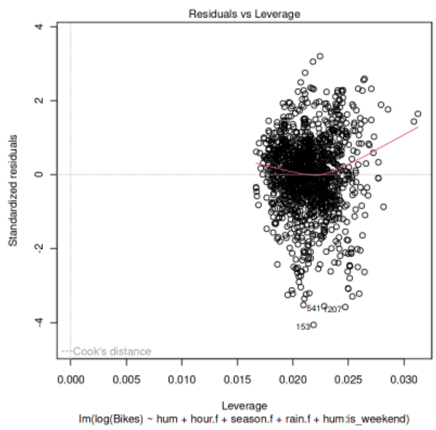
Scale-Location Plot:

The red line is not perfectly horizontal, however it doesn't deviate wildly at any point. Thus we can declare that the scale-location plot does not provide much evidence against the assumption of equal variance.



Residuals vs Leverage Plot:

This plot is used to identify influential observations, as none of the points lie outside of Cook's distance, (the dashed lines do not even appear on the plot!) there are no influential outliers and outlier removal is not necessary.



Conclusion: With some trepidation due to the QQ plot, we will follow the evidence provided by the Residuals vs Fitted plot and Scale location plot, which is in favour of the assumption that the residuals are normally distributed after a log transform.

Interpretation of the Coefficients:

The output for the summary of our model is found below:

```
Call:
lm(formula = log(Bikes) ~ hum + hour.f + season.f + rain.f +
    hum:is_weekend, data = train)

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     6.262796   0.130320  48.057  < 2e-16 ***
hum            -0.010545   0.001474  -7.155 1.36e-12 ***
hour.f1        -0.229728   0.103169  -2.227 0.026128 *
hour.f2        -0.886888   0.098425  -9.011  < 2e-16 ***
hour.f3        -1.210337   0.104379 -11.596  < 2e-16 ***
hour.f4        -1.300846   0.101100 -12.867  < 2e-16 ***
hour.f5        -0.719296   0.100774  -7.138 1.53e-12 ***
hour.f6         0.667457   0.096222   6.937 6.17e-12 ***
hour.f7         1.266482   0.105007  12.061  < 2e-16 ***
hour.f8         2.134565   0.098018  21.777  < 2e-16 ***
hour.f9         1.895642   0.101499  18.677  < 2e-16 ***
hour.f10        1.396006   0.100267  13.923  < 2e-16 ***
hour.f11        1.385973   0.106119  13.061  < 2e-16 ***
hour.f12        1.518430   0.104042  14.594  < 2e-16 ***
hour.f13        1.619571   0.098470  16.447  < 2e-16 ***
hour.f14        1.618510   0.104263  15.523  < 2e-16 ***
hour.f15        1.692418   0.102711  16.477  < 2e-16 ***
hour.f16        1.895197   0.103737  18.269  < 2e-16 ***
hour.f17        2.098072   0.103765  20.219  < 2e-16 ***
hour.f18        2.043661   0.099105  20.621  < 2e-16 ***
hour.f19        1.719117   0.099400  17.295  < 2e-16 ***
hour.f20        1.333224   0.102270  13.036  < 2e-16 ***
hour.f21        1.010828   0.100432  10.065  < 2e-16 ***
hour.f22        0.862057   0.101155   8.522  < 2e-16 ***
hour.f23        0.492430   0.100632   4.893 1.11e-06 ***
season.fSpring  0.328702   0.041786   7.866 7.36e-15 ***
season.fSummer  0.223364   0.042467   5.260 1.67e-07 ***
season.fAutumn -0.096222   0.043988  -2.187 0.028876 *
rain.fRaining  -0.190344   0.033766  -5.637 2.10e-08 ***
hum:is_weekend -0.001600   0.000453  -3.532 0.000426 ***
```

The model has reference categories hour.f = 0, season.f = December, and rain.f = Not Raining, is_weekend = No.

- **Hum:** The coefficient multiplying humidity indicates that for every unit increase in humidity the number of bikes hired decreases by 1.06%.
- **hour.f:** With many levels in the hour.f variable, providing an interpretation for each one is impractical. In general, the hour.f variable adjusts the intercept by its corresponding coefficient for each level, relative to the reference category (hour.f = 0).
- **season.f:** The reference category here is December, relative to December the amount of bikes hired increases by 33% in the Spring, 22% in the Summer, and decreases by 10% in Autumn.
- **rain.fRaining:** The reference category here is when it's not raining, so relative to the reference category the number of bikes hired decreases by 19% when it is raining.
- **hum:is_weekend:** There is no additional effect on humidity when it is a weekday, however on the weekend there is a 1.06% + 0.16% = 1.22% decrease in bikes hired on the weekend per unit of humidity.

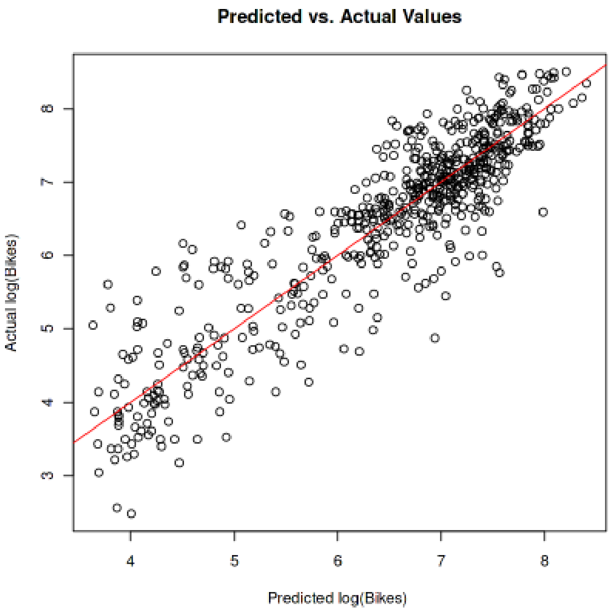
Interpretation of Summary output and Assessment on Testing Dataset:

```
Residual standard error: 0.5557 on 1370 degrees of freedom
Multiple R-squared:  0.8137,    Adjusted R-squared:  0.8098
F-statistic: 206.3 on 29 and 1370 DF,  p-value: < 2.2e-16
```

The R^2 of 0.8137 indicates that the model explains 81.37% of the variability in the number of bikes hired. The Adjusted R^2 of 0.8098 is very close to the actual R^2, indicating that we are likely not overfitting the model.

The residual standard error of 0.5557 is the average amount by which the log(Bikes) will deviate from the true regression line.

This plot is using the testing dataset and indicates that the model performs quite well. The red line tracks the points in a linear fashion and the mean squared error was calculated to be 0.3009222. This indicates that on average the prediction for log(Bikes) was off by 0.3009222 units of log(Bikes).



Part B) Logistic Regression

1. The fitted logistic regression model is as follows,

```
Call:
glm(formula = Attrition.f ~ BusinessTravel.f + Department.f +
    Gender.f + JobSatisfaction.f + EmployeeID + HourlyRate +
    DistanceFromHome + Age, family = binomial, data = turnover)
```

The odds ratios and 95% confidence intervals are as follows;

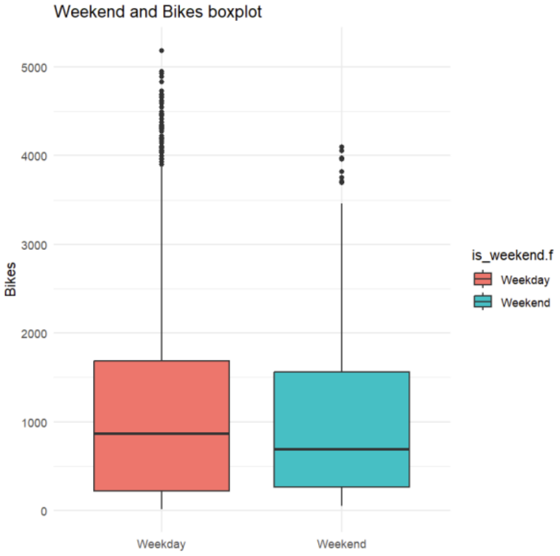
- **Travel Rarely:** 0.4984013 with 95% confidence interval (0.3577068, 0.6984227)
    - This indicates that employees in the 'travel rarely' category have approximately 50% lower odds of leaving their job (attrition) compared to employees who 'travel frequently' (the reference category), holding all other variables constant.
- **No Travel:** 0.2319109 with 95% confidence interval (0.1140160, 0.4369779)
    - This indicates that employees in the 'no travel' category have approximately 77% lower odds of leaving their job compared to employees in the reference category 'travel frequently', holding all other variables constant.
- **Medium Job Satisfaction:** 0.6636597 with 95% confidence interval (0.4676707, 0.9470045)
    - This indicates that employees in the 'medium job satisfaction' category have approximately 34% lower odds of leaving their jobs compared to employees in the reference category 'Low Satisfaction', holding all other variables constant.
- **High Job Satisfaction:** 0.3892953 with 95% confidence interval (0.2559370, 0.5887910)
    - This indicates that employees in the 'high job satisfaction' category have approximately 61% lower odds of leaving their jobs compared to employees in the reference category 'Low Satisfaction', holding all other variables constant.
- **Distance:** 1.0284939 with 95% confidence interval (1.00108875, 1.0461796)
    - This indicates that for every 1km increase in distance from home to work, the odds of attrition increase by approximately 2.85%, holding all other variables constant.
- **Age:** 0.9467307 with 95% confidence interval (0.9308185, 0.9641738)
    - This indicates that for every year increase in age, the odds of attrition decrease by approximately 5.33%.

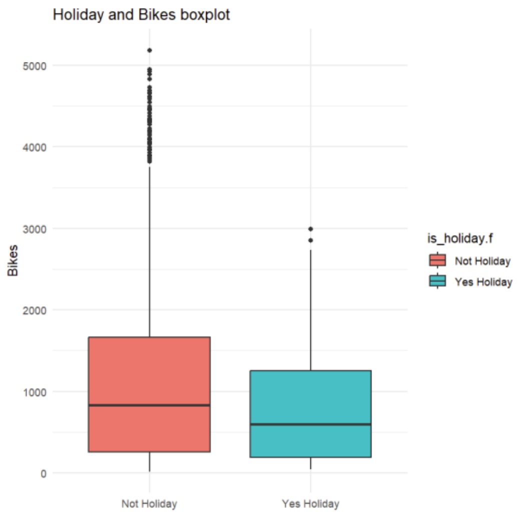
Some conclusions we can draw from this analysis are:

- Employees don't like to travel – the more an employee travels the higher their odds of attrition are.
- Job satisfaction matters – employees with medium job satisfaction have lower odds of attrition than those with low job satisfaction, and employees with high job satisfaction have lower odds of attrition than those with medium job satisfaction.
- Employees don't like to commute – the further an employee lives from work the higher their odds of attrition are.
- Loyalty grows with age – Older employees have lower odds of attrition compared to younger employees.

Appendix A :

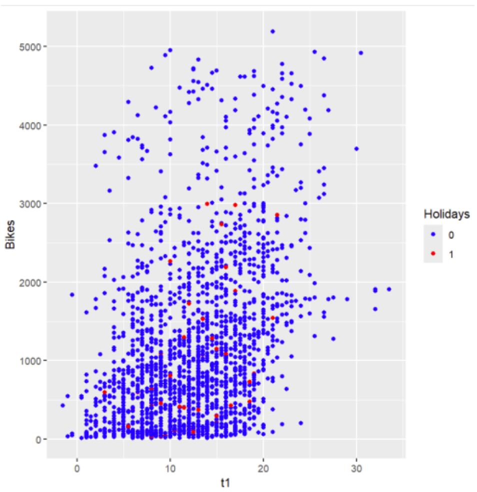
Weather , holiday and bikes boxplots





There does appear to be a bit of a relationship between holiday and bikes hired but there were so few data points for holidays in my dataset that I thought it best to err on the side of caution and exclude the variable from my model.

Is_holiday:t1 interaction plot



Vif values for the second linear model:

	Est.	S.E.	t val.	p	VIF
(Intercept)	868.06	161.39	5.38	0.00	
hum	-9.70	1.68	-5.76	0.00	2.17
season.fSpring	89.74	58.35	1.54	0.12	2.55
season.fSummer	94.33	48.72	1.94	0.05	2.55
season.fAutumn	-0.17	48.85	-0.00	1.00	2.55
rain.fRaining	-187.16	37.13	-5.04	0.00	1.29
hour.f1	54.41	112.18	0.49	0.63	1.79
hour.f2	-96.74	106.97	-0.90	0.37	1.79
hour.f3	-154.54	113.48	-1.36	0.17	1.79
hour.f4	-99.84	109.93	-0.91	0.36	1.79
hour.f5	-161.90	109.47	-1.48	0.14	1.79
hour.f6	307.55	104.65	2.94	0.00	1.79
hour.f7	1131.58	114.14	9.91	0.00	1.79
hour.f8	2661.51	106.48	25.00	0.00	1.79
hour.f9	1397.47	110.34	12.66	0.00	1.79
hour.f10	635.21	109.09	5.82	0.00	1.79
hour.f11	661.76	115.34	5.74	0.00	1.79
hour.f12	829.21	113.11	7.33	0.00	1.79
hour.f13	969.13	107.12	9.05	0.00	1.79
hour.f14	926.03	113.63	8.15	0.00	1.79
hour.f15	1053.59	112.00	9.41	0.00	1.79
hour.f16	1357.63	112.91	12.02	0.00	1.79
hour.f17	2121.67	113.07	18.76	0.00	1.79
hour.f18	1989.61	107.78	18.46	0.00	1.79
hour.f19	1173.13	108.09	10.85	0.00	1.79
hour.f20	628.79	111.28	5.65	0.00	1.79
hour.f21	388.79	109.20	3.56	0.00	1.79
hour.f22	276.96	110.08	2.52	0.01	1.79
hour.f23	114.01	109.36	1.04	0.30	1.79
hum:is_weekend.fWeekend	-9.02	0.96	-9.39	0.00	4.00
is_weekend.fWeekday:t1	25.32	5.03	5.03	0.00	10.06
is_weekend.fWeekend:t1	59.04	5.81	10.17	0.00	10.06

The code used to fit the final model was

```
model <- lm(Bikes ~ hum + season.f + rain.f + hour.f + is_weekend.f:hum, data = train)
```

Appendix B:

Code for fitting the logistic regression model.

```
glm_model <- glm(Attrition.f ~ BusinessTravel.f + Department.f + Gender.f +
                 + JobSatisfaction.f + EmployeeID + HourlyRate + DistanceFromHome +
                 + Age, data=turnover, family = binomial)
```